

DR BEKDAULET SHUKIRGALIYEV



PERSONAL INFORMATION

Born in Kazakhstan, 21 April 1987
married, four sons and a daughter

nationality Republic of Kazakhstan

current position Instructor at Nazarbayev University

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Web of Science N-4025-2014 (h-index=10)
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Scopus 57163629900 (h-index=10)
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Google Scholar (h-index=12)

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EDUCATION

Doctor of Science in Astronomy

09/2015-11/2018 Heidelberg University, (Heidelberg, Germany)
Cum laude · School: International Max Planck Research School for Astronomy and Cosmic Physics (IMPRS-HD)/Heidelberg Graduate School of Fundamental Physics (HGSFP)
Thesis: *The life of star clusters, from birth to dissolution: a new approach*
Advisors: Priv.-Doz. Dr. Geneviève PARMENTIER & Prof. Andreas JUST

Master of Natural Sciences

09/2011-07/2013 Al-Farabi Kazakh National University, (Almaty, Kazakhstan)
GPA: 3.89/4.0 · Specialty: *Astronomy* · Faculty of Physics and Technology
Thesis: *Dynamical evolution of variable-mass binary system in the presence of reactive forces*
Advisors: Prof. Mukhtar MINGLIBAYEV & Dr. Aisha NAURYZBAYEVA

Bachelor of Natural Sciences

09/2006-06/2010 Al-Farabi Kazakh National University, (Almaty, Kazakhstan)
GPA: 3.67/4.0 · Specialty: *Astronomy* · Faculty of Physics
Diploma work: *The dynamics of binary stars*
Advisors: Prof. Mukhtar MINGLIBAYEV, Prof. Zeinulla ZHANABAYEV & Ms. Farida BAIDOLDA

Vocational Diploma

09/2002-07/2006 Uralsk college of gas, oil and branch technology, (Uralsk, Kazakhstan)
GPA: 5.0/5.0 · Specialization: *software engineering for computers and computer-based systems* · Construction and technical faculty
Diploma work: *Plotting mathematical functions* Description: I made my own software which can plot the graph of any mathematical function reading from the text. The software was built on the Borland Delphi environment for Windows-based systems.

Middle school certificate

09/1993-07/2002 Zhangala secondary school (Zhangala, Kazakhstan)
GPA: 5.0/5.0

WORK (RESEARCH/TECHNICAL) EXPERIENCE

since Jan 2025	<i>Instructor</i>	Nazarbayev University, Physics Department (Astana, Kazakhstan)	I teach Physics courses, including Physics I and II, Optics, Modern Physics, and Scientific Parallel Computing, including lectures, tutorials, and lab work. Additionally, I am a laboratory supervisor for Physics lab courses, where I am reviewing and improving lab experiments, manuals, and equipment. I am also conducting research in Computational Astrophysics and am the principal investigator of a grant project funded by the Ministry of Science and Higher Education. Reference: Head of Department, Anton DESYATNIKOV · anton.desyatnikov@nu.edu.kz
Oct 2024 – Jan 2025	<i>Project Leader (Dean)</i>	Heriot-Watt University Aktobe Campus (Aktobe, Kazakhstan)	As Dean, I oversee academic and administrative operations at the campus, ensuring alignment with institutional goals. I am responsible for communication between administrative units at Heriot-Watt University in Edinburgh and Aktobe, and I manage the delivery of materials and resources from Edinburgh. My research continues in Computational Astrophysics as the principal investigator of a grant project funded by the Ministry of Science and Higher Education. Reference: Rector, Laura KARABASSOVA · lkarabassova@zhubanov.edu.kz
Jan 2024 – Oct 2024	<i>Associate Professor</i>	Heriot-Watt University Aktobe Campus (Aktobe, Kazakhstan)	I teach Physics courses to first-year Bachelor students, including lectures, tutorials, and lab work. I am also conducting research in Computational Astrophysics and am the principal investigator of a grant project funded by the Ministry of Science and Higher Education. Reference: Dean, Galymzhan TULEUSHOV · G.Tuleushov@hw.ac.uk
since May 2024	<i>Research Assistant</i>	Nazarbayev University, Physics Department (Astana, Kazakhstan)	Participating in a grant project funded by the Ministry of Science and Higher Education [AP13067834] “Understanding the properties of star clusters with artificial intelligence”. Reference: Project PI, Dana ALINA · dana.alina@nu.edu.kz
Sep 2023 – June 2024 (part-time)	<i>Associate Professor</i>	Astana IT University (Astana, Kazakhstan)	Taught Machine Learning Algorithms (lecture and practice) to second and third-year undergraduate students. Also led the full course on High-Performance Computing for first-year graduate students. Reference: Head of Department of Computation and Data Science, Dr. Muslim SERGAZIYEV · muslim.sergaziyev@astanait.edu.kz
Jan 2024 – Apr 2024 (part-time)	<i>Postdoctoral Scholar</i>	Nazarbayev University, Energetic Cosmos Laboratory, (Astana, Kazakhstan)	A key member of one faculty development grant at NU. Research on the formation and evolution of Open Clusters and the dynamical evolution of Active galactic nuclei using massively parallel high-performance computing. Massively parallel N -body simulations on GPUs and computer simulations of core-collapse Super Novae explosion processes. · Performing data analysis on simulations of large data sets. Optimization and parallelization of data-analysis programming tools. Reference: Associate director, Prof. Ernazar ABDIKAMALOV · ernazar.abdikamalov@nu.edu.kz
Apr 2019–Dec 2023	<i>Postdoctoral Scholar</i>	Nazarbayev University, Energetic Cosmos Laboratory, (Astana, Kazakhstan)	PI of the two research projects (MES grants) and a key member of one Faculty development grant of NU. Research on the formation and evolution of Open Clusters and the dynamical evolution of Active galactic nuclei using massively parallel high-performance computing. Massively parallel N -body simulations on GPUs and computer simulations of core-collapse

Super Novae explosion processes. · Performing data analysis on simulations of large data sets. Optimization and parallelization of data-analysis programming tools. · Building and administering local HPC workstations with powerful GPUs (RTX3090).
Reference: Associate director, Prof. Ernazar ABDIKAMALOV · ernazar.abdikamalov@nu.edu.kz

Senior researcher Fesenkov Astrophysical Institute
(Almaty, Kazakhstan)

Apr 2019 – Feb 2024
(part-time)

Research on star-disk interactions in active galactic nuclei using massively parallel N -body simulations. · Participating in the development of the local HPC-facilities
Reference: Director, Dr. Chingis OMAROV · omarov@aphi.kz

Senior lecturer Al-Farabi Kazakh National University,
Faculty of Physics and Technology, (Almaty, Kazakhstan)

Sep 2020–Jun 2021
(part-time/online)

I taught a full course of ‘Stellar dynamics.’ Supervised three students on computational astrophysics research: a last-year undergraduate student, one master’s student, and one Ph.D. student.

Reference: Head of department, Dr. Margulan IBRAIMOV · Margulan.Ibraimov@kaznu.kz

Postdoctoral researcher Astronomisches Rechen-Institut at
Zentrum für Astronomie der Universität Heidelberg (Heidelberg, Germany)

Jan 2019–Mar 2019

Fellowship from Volkswagen Trilateral Grant for the research on the field of dynamics of the nuclear cluster in the active galactic nuclei. Testing the performance of the updated N -body code for AGN simulations.

Reference: Prof. Rainer SPURZEM · spurzem@ari.uni-heidelberg.de

Researcher Astronomisches Rechen-Institut at
Zentrum für Astronomie der Universität Heidelberg (Heidelberg, Germany)

Sep 2015–Dec 2018
(part-time)

Studied the evolution of star clusters from their birth till their dissolution in the Galactic tidal field by means of direct N -body simulations. · Performed about 1000 massively parallel N -body simulations of star cluster evolution with special initial conditions. · Performed big-data analysis of N -body simulations

Reference: Dr. Geneviève PARMENTIER · gparm@ari.uni-heidelberg.de

Visiting researcher National Astronomical Observatories of China,
Chinese Academy of Science (Beijing, China)

Nov-Dec, 2013

I analyzed the outputs of the large simulation to study the effect of star-disc interactions on the distribution of orbital parameters of stars accreted by a super-massive black hole in AGN.

Reference: Prof. Rainer SPURZEM · spurzem@nao.cas.cn

Assistant Teacher Al-Farabi Kazakh National University,
Faculty of Physics and Technology (Almaty, Kazakhstan)

Sep 2013–Oct 2014
(part-time)

I tutored practical and laboratory classes of special courses to undergraduate students such as ‘Stellar Structure and Evolution,’ ‘Dynamics of Planetary Systems,’ ‘Fractals in the Universe,’ and ‘Analog electronic circuits.’

Reference: Head of department, Dr. Margulan IBRAIMOV · Margulan.Ibraimov@kaznu.kz

Junior researcher Fesenkov Astrophysical Institute
(Almaty, Kazakhstan)

Aug 2013–Mar 2019

Studied the star-disc interactions in Active Galactic Nuclei in collaboration with Prof. Dr. Rainer Spurzem (Germany, China). Used several foreign large HPC facilities to perform direct N -body simulations and extensive data analysis of large simulation outputs. Developed data analysis tools on C/C++ combined with GNUPLOT via BASH-scripts.

Reference: Director, Dr. Chingis OMAROV · chingis.omarov@gmail.com

Visiting researcher National Astronomical Observatories of China,
Chinese Academy of Science (Beijing, China)

Nov-Dec, 2012

I studied the effect of star-disc interactions on the distribution of orbital parameters of stars accreted by a super-massive black hole in AGN. Learned to write BASH-scripts and to make

plots with GNUPLOT remotely on Linux-based systems. Made first N -body simulations on phi-GRAPE/GPU (CUDA/C) and nbbody6++ (CUDA/FORTRAN) codes.

Reference: Prof. Rainer SPURZEM · spurzem@nao.cas.cn

Research Engineer

Fesenkov Astrophysical Institute
(Almaty, Kazakhstan)

Jan 2011–July 2013
Feb 2010–Aug 2010

Studied star-disc interactions in AGN as a part of Germany-Kazakhstan Collaboration Project STARDISK funded by Volkswagen Stiftung (Germany). Ran several N -body simulations on large- and mid-size supercomputers MilkyWay (JSC FZ-Jülich, Germany), Laohu (NAOC, China), Kepler (former Titan at Uni-Heidelberg, Germany). Studied the variable-mass two-body problem analytically in case of the existence of reactive forces.

Reference: Director, Dr. Chingis OMAROV · chingis.omarov@gmail.com

IT-specialist

West-Kazakhstan Regional Court
(Uralsk, Kazakhstan)

Feb 2006–Aug 2006

Functioned as System Administrator, administrator of the electronic database of Court documents, general IT-support. Extended and set up the local network (LAN) in the building of the Court.

Reference: HR-department +7 (7112) 55-47-63 · 0701@sud.kz

IT-support trainee

Karachaganak Petroleum Operating
(Aksai, Kazakhstan)

Jul 2005–Aug 2005

One month paid internship from KPO. I worked in IT support team. Duties: Fixing software and hardware problems, replacing peripheral hardware, Installation and first setup of Windows XP operating system on new PCs, installation of new PCs at places, and setting up the network connection.

TEACHING EXPERIENCE

NAZARBAYEV UNIVERSITY, PHYSICS DEPARTMENT — Astana, Kazakhstan

Spring, 2026

“Optics” lab sessions. “Physics II” Recitations, four sections.

Reference: Head of Department, Anton DESYATNIKOV · anton.desyatnikov@nu.edu.kz

Fall, 2025

“Scientific Parallel Computing” half-course, including lectures and practice classes for third and fourth-year undergraduate and graduate students. “Modern Physics” lab sessions. “Physics I” Recitations, four sections.

Reference: Head of Department, Anton DESYATNIKOV · anton.desyatnikov@nu.edu.kz

Summer, 2025
Recitation

“Physics I and II” Recitation sessions.

Reference: Head of Department, Anton DESYATNIKOV · anton.desyatnikov@nu.edu.kz

Spring, 2025
Lab courses

“Optoelectronics”, “Optics”, “Physics I” Laboratory sessions.

Reference: Head of Department, Anton DESYATNIKOV · anton.desyatnikov@nu.edu.kz

HERIOT-WATT UNIVERSITY AKTOBE CAMPUS — Aktobe, Kazakhstan

Trimesters 1,2 and 3,
2024
Full course

“Physics” course for first-year undergraduate students. Lecture+Tutorial $(2 + 2) \times 5 = 20$ contact hours a week for two trimesters and Lab course (25 contact hours a week) for one semester

Reference: Galymzhan TULEUSHOV · G.Tuleushov@hw.ac.uk

ASTANA IT UNIVERSITY, DEPARTMENT OF COMPUTATION AND DATA SCIENCE —
Astana, Kazakhstan

Trimester 1, 2023-2024 Practice classes “Machine Learning Algorithms” course for second-year undergraduate students (3 groups, three contact hours a week each) taught online via Microsoft teams.
Reference: Dr. Muslim SERGAZIYEV · muslim.sergaziyev@astanait.edu.kz

Trimester 2, 2023-2024 full course “Machine Learning Algorithms” course for third-year undergraduate students. Two contact hours of lectures and nine contact hours of practice classes a week taught online via Microsoft Teams.
Reference: Dr. Muslim SERGAZIYEV · muslim.sergaziyev@astanait.edu.kz

Trimester 3, 2024 full course “High-Performance Computing” course for first-year graduate students. Five contact hours in total taught online via Microsoft Teams.
Reference: Dr. Muslim SERGAZIYEV · muslim.sergaziyev@astanait.edu.kz

AL-FARABI KAZAKH NATIONAL UNIVERSITY, FACULTY OF PHYSICS AND TECHNOLOGY, DEPARTMENT OF SOLID STATE AND NON-LINEAR PHYSICS — Almaty, Kazakhstan

WS 2020 Full course “Stellar Dynamics” special course for undergraduate students. Lecture + practical (1+2) classes (fully on-line)
SS 2014 Practical classes “Stellar structure and evolution” and “Fractals in the Universe” special courses for undergraduate students. Practical classes
WS 2013 Laboratory classes “Analogue electronic circuits” and “Dynamics of planetary systems” special courses for undergraduate students. Laboratory classes.
Reference: Dr. Margulan IBRAIMOV · Margulan.Ibraimov@kaznu.kz

HEIDELBERG UNIVERSITY, THE NATURAL SCIENCES AND MATHEMATICS FACULTY — Heidelberg, Germany

SS 2016 Tutorial “Introduction to Computational Physics” course for undergraduate and graduate students. Tutorial classes.
Reference: Prof. Ralf KLESSEN · klessen@uni-heidelberg.de and Prof. Rainer SPURZEM · spurzem@ari.uni-heidelberg.de

SKILLS

Tools and Languages C/C++, PYTHON, FORTRAN, PASCAL, Git, L^AT_EX, HTML, PHP, BASH-scripting

Parallel programming CUDA/C, MPI and OpenMP

Programs/codes used in research MATLAB, Mathematica, Open source *N*-body codes: phi-GRAPE/GPU, NBODY6++GPU, PeTar, MKHALO (falcon), galaxia, magnetohydrodynamics code FLASH, scientific packages for PYTHON: SciPy, NumPy, CuPy, Pandas, AstroPy, GalPy, CuNumeric

GPU-accelerated HPC clusters I have used MILKYWAY, JURECA, JUWELS at Juelich Supercomputing Center (Jülich, Germany) · HYDRA, KEPLER, BWForCluster(MLS&WISO) at Heidelberg University (Heidelberg/Mannheim, Germany) · LAOHU at National Astronomical Observatories of China (Beijing, China) · SHABYT, DGX-1, DGX-2 at Nazarbayev University (Astana, Kazakhstan) · smaller HPC systems at Fesenkov Astrophysical Institute (Almaty, Kazakhstan) and Energetic Cosmos Laboratory of Nazarbayev University (Astana, Kazakhstan)

Additional IT-skills I have experience installing, setting up, and system administrating Linux and Windows-based systems, building and administering local networks, administering databases, and assembling hardware.

Communication KAZAKH (Native) · ENGLISH (Fluent) · RUSSIAN (Fluent) · TURKISH (Pre-intermediate) · GERMAN (Elementary) · ARABIC (Beginner)

FUNDING, AWARDS AND GRANTS

<i>State prize</i>	Dec 2024 · Best University Teacher
<i>Research funding as Principal Investigator</i>	Jul 2023–Dec 2025 · Research Grant [AP19677351] “Studying the formation of star clusters with combined magnetohydrodynamics and N-body simulations” from the Ministry of Education and Science of the Republic of Kazakhstan. Funds: 99 232 339.02 KZT or about 220 000 USD. (ECL/NU, Astana, Kazakhstan)
<i>Research funding as Principal Investigator</i>	Apr 2022–Dec 2024 · Research Grant for young researchers [AP13067834] “Understanding the properties of star clusters with artificial intelligence” from the Ministry of Education and Science of the Republic of Kazakhstan. Funds: 65 944 678.65 KZT or about 155 000 USD. (ECL/NU, Astana, Kazakhstan)
<i>Research funding as Key participant</i>	Jan 2022–Dec 2024 · Nazarbayev University Faculty Development Competitive Research Grant Program No 11022021FD2912 “Unveiling stellar physics, from birth to explosion, using AI”. PI: Prof. Ernazar Abdikamalov & Prof. Dana Alina (SSH/NU, Astana, Kazakhstan)
<i>Research funding as Principal Investigator</i>	Oct 2020–Dec 2022 · Research Grant [AP08856149] “Star Cluster dissolution mechanisms in the Milky-Way-like galaxies” from the Ministry of Education and Science of the Republic of Kazakhstan. Funds: 67 352 300, 90 KZT or about 160 000 USD. (ECL/NU, Astana, Kazakhstan)
<i>Travel grant</i>	May 2019 · Travel-grant from International Astronomical Union for participation in IAU Symposium 351 (Bologna, Italy)
<i>Travel grant</i>	Mar 2019 · Travel-grant from Simons Foundation for participation in Black Holes in the Disks of Active Galactic Nuclei Workshop (Flatiron Institute, New York city, NY, USA)
<i>Research funding as team member</i>	Jan-Mar 2019 · Fellowship from Volkswagen Trilateral Grant (PI: Rainer Spurzem, Heidelberg University, Germany)
<i>Funding for collaborative research as key team member</i>	2018–2020 · Funding for collaboration from the International Partnership Program of Chinese Academy of Sciences, Grant No. 114A11KYSB20170015 (GJHZ1810). (PIs: Prof. Rainer Spurzem and Dr. Chingis Omarov)
<i>Stipend</i>	Sep 2018 – Aug 2019 · State Scientific Stipend for young researchers from The Ministry of Education and Science of The Republic of Kazakhstan about 1900 USD in total. (Fesenkov Astrophysical Institute, Almaty, Kazakhstan)
<i>Research funding as team member</i>	Sep 2015–Dec 2018 · Fellowship from SFB-881 “The Milky Way System” sub-project B2 funded by DFG (PI: Genevieve Parmentier, Heidelberg University, Germany)
<i>PhD Fellowship</i>	Sep 2015–Dec 2018 · Fellowship from International Max Planck Research School for Astronomy and Cosmic Physics (Max Planck Institute for Astronomy, Heidelberg, Germany)
<i>Travel grant</i>	Aug 2014 · Travel-grant from IAU for participation in IAU Symposium 312 (Beijing, China)
<i>Stipend</i>	Sep 2011–Jun 2013 · State Grant stipend for education, Ministry of Education and Science of Kazakhstan
<i>Stipend</i>	Sep 2006–Jun 2010 · State Grant stipend for education, Ministry of Education and Science of Kazakhstan
<i>Stipend</i>	Sep 2002–Jun 2006 · Regional Grant stipend for education, West-Kazakhstan regional Mayor’s office

MEMBERSHIP IN PROFESSIONAL SOCIETIES AND COMMITTEES

<i>Since Jan 2021</i>	Individual member of International Astronomical Union / www.iau.org
<i>Since Feb 2019</i>	Ordinary member of European Astronomical Society / eas.unige.ch
<i>Since 2018</i>	Member of Young Researchers Alliance (Kazakhstan) / yrascience.kz
<i>Since 2025</i>	Member of (Kazakhstan) / yrascience.kz

CONFERENCES AND WORKSHOPS ATTENDED

May 2026 oral presentation	<i>A Decade of Centrally Concentrated Cluster Models: Formation, Survival, and Observations</i> at the 2nd Meeting of The Silk Road Astronomy Research and Education Alliance, May 18-22, 2026, Tashkent, Uzbekistan
June 2025 oral presentation	<i>Simulating Centrally Concentrated Star Formation with MHD and N-body Models</i> at the 1st Meeting of The Silk Road Astronomy Research and Education Alliance, June 23-26, 2025, Dali, China
May 2024 special talk	<i>Stellar-feedback-driven gas expulsion in star clusters formed with centrally-peaked efficiency</i> at the 12th Kazakhstan-China-Korea (KCK) Workshop, May 20-24, 2024, Astana, Kazakhstan
July 2023 poster	<i>How can Open Clusters have stellar mass black holes?</i> at the European Astronomical Society Annual Meeting (EAS 2023), July 10-14, 2023, Krakow, Poland
June 2023 invited speaker	<i>Evolution of star clusters formed with the centrally-peaked star formation efficiency</i> at 25th Meeting on Research in Astronomy IASBS, June 14-16, 2023, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran (ONLINE)
January 2023 oral presentation	<i>The impact of the Galactic tidal field on the dissolution time of star clusters</i> at Timescales in Astrophysics, January 15-21, 2023, New York University Abu Dhabi, Abu Dhabi, UAE
November 2022 oral presentation	<i>High-performance computing and AI.</i> at HPC & AI at NU Symposium, November 28, 2022, Astana, Kazakhstan
June 2022 poster	<i>Evolution of star clusters with centrally peaked star formation</i> at From Stars to Galaxies II - Connecting our understanding of star and galaxy formation, June 20-24, 2022, Chalmers University of Technology, Gothenburg, Sweden
December 2021 oral presentation	<i>Massively parallel direct N-body simulations.</i> at NU Symposium on HPC & AI, December 8, 2021, Nazarbayev University, Astana, Kazakhstan (ONLINE)
September 2021 oral presentation	<i>The evolution star clusters with centrally peaked star-formation efficiency.</i> at Volkswagen Trilateral Project 3rd Meeting on Dynamical Mechanisms of Accretion in Galactic Nuclei (2021), September 6-8, 2021, Heidelberg University, Heidelberg, Germany (ONLINE)
May 2021 oral presentation	<i>The evolution star clusters with centrally peaked star-formation efficiency.</i> at The International Online Workshop on Relativistic Astrophysics and Gravitation IWRAG-2021, May 12-14, 2021, Tashkent, Uzbekistan (ONLINE)
December 2020 oral presentation	<i>Violent relaxation and survivability of Star Clusters with centrally peaked SFE-profiles.</i> at The 2020 Cross-Strait Star Cluster online conference, December 7-8, 2020, Suzhou, China (ONLINE)
December 2019 oral presentation	<i>The life of bound clusters from the formation till dissolution</i> at 11th Korea-China-Kazakhstan (KCK) Meeting on Stellar Dynamics and Gravitational-Wave Astronomy, December 12-13, 2019, Daejeon, Republic of Korea
September 2019 oral presentation	<i>Studying the evolution of star clusters from birth to dissolution with a new approach</i> at CoCoNuT Meeting 2019, September 18-20, 2019 Astana, Kazakhstan
June 2019 oral presentation	<i>Star-disk interactions in Active Galactic Nuclei</i> at ECL19: Exploring the Energetic Universe, June 17-21, 2019 Astana, Kazakhstan
May 2019 poster	<i>Star Clusters in the Galactic Tidal Field, from birth to dissolution</i> at Star Clusters: from the Milky Way to the Early Universe, IAU Symposium 351 & MODEST-19, May 27-31, 2019 Bologna, Italy

March 2019 oral presentation /discussion	<i>STARDISK Project at the Black Holes in the Disks of Active Galactic Nuclei Workshop, March 11-13, 2019, New York, NY, USA</i>
November 2018 oral presentation	<i>The life of star clusters from formation to dissolution: a novel approach at Survival of Dense Star Clusters in the Milky Way System, Heidelberg, Germany</i>
August 2018 oral presentation	<i>Star clusters formed with centrally peaked SFE profile at International Workshop Star Clusters around the Milky Way and in the Local Group (Observations, Dynamics, Modeling, Super-computing), Heidelberg, Germany</i>
May 2018 poster	<i>The dissolution of star clusters formed with centrally peaked SFE profile at Olympian Symposium 2018, Paralia, Greece</i>
June 2017 poster	<i>The impact of a star-formation efficiency profile on the evolution of open clusters at Francesco's Legacy: Star Formation in Space and Time, Florence, Italy</i>
April 2017 oral presentation	<i>The impact of a star-formation efficiency profile on the evolution of open clusters at The Origin and Evolution of the Milky Way System, SFB881 Retreat, Kloster Schöntal, Germany</i>
September 2016 oral presentation	<i>Improving the Survival Likelihood of Star Clusters After Instantaneous Gas Expulsion at MODEST-16 NYC: A Conference on Gas and Gravitational Dynamics, New York City, NY, USA</i>
August 2016 oral presentation	<i>Improving the Survival Likelihood of Star Clusters After Instantaneous Gas Expulsion at Star Clusters: From Infancy to Teenagehood, Heidelberg, Germany</i>
October 2015 oral presentation	<i>Initial conditions for modeling multi-component systems at International Workshop Advances in Space Research and Technologies, Almaty, Kazakhstan</i>
December 2014 oral presentation	<i>Star-disk interactions in AGN: The effect on compact stellar cluster at Korea-China-Kazakhstan Meeting on Stellar Dynamics, Seoul, Korea</i>
August 2014 oral presentation	<i>The effect of gaseous accretion disk on dynamics of the stellar cluster in AGN at International Astronomical Union Symposium IAUS-312: Star Clusters and Black Holes in Galaxies across cosmic time, Beijing, China</i>
July 2013 oral presentation	<i>Capturing stars into massive black holes via accretion disks at European Week of Astronomy and Space Science (EWASS-2013), Turku, Finland</i>

CONFERENCE ORGANIZATION

June 2025 SOC member	The 1st Meeting of The Silk Road Astronomy Research and Education Alliance, Dali University, Dali, China
May 2024 SOC and LOC-chair	The 12th Kazakhstan-China-Korea (KCK) Workshop, Nazarbayev University, Astana, Kazakhstan (Chair of the first session)
November 2022 LOC member	HPC/AI @ NU '22 Symposium, Nazarbayev University, Astana, Kazakhstan (Chair of the afternoon session)
December 2021 LOC member	NU Symposium on HPC & AI, Nazarbayev University, Astana, Kazakhstan (Chair of the afternoon session)
August 2017 LOC member	Star Clusters: From Infancy to Teenagehood, Heidelberg University, Heidelberg, Germany
August 2013 LOC member	MODEST-13: Star Clusters across cosmic time, Fesenkov Astrophysical Institute, Almaty, Kazakhstan

COLLOQUIUM AND SEMINARS

- 14 November 2024 *Resilient Star Clusters: Surviving Gas Loss with Centrally Peaked Star Formation*
AMNH Astro Seminar, Department of Astrophysics at the American Museum of Natural History, New York, NY, USA
- 9 November 2023 *Star clusters with centrally-peaked star-formation efficiency*
Physics Seminar, School of Mathematics and Physics, Xi'an Jiaotong Liverpool University (XJTLU) at Suzhou, China
- 28 February 2023 *From star clusters to ChatGPT*
Open Astrophysics Seminar, Nazarbayev University, Astana, Kazakhstan
- 5 April 2022 *Black holes in Open Clusters after violent relaxation*
Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany (ONLINE)
- 22 April 2020 *Dehnen models with centrally peaked SFE-profiles*
Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany (ONLINE)
- 1 November 2019 *Star Clusters in the Galactic tidal field: birth, death and everything in between*
Teeminar (the group seminar of Silk Road Project), National Astronomical Observatory of China, Beijing, China
- 30 May 2019 *Fast Radio Bursts (FRB): What are they?*
Research Seminar at Physics Department, School of Science and Technology of Nazarbayev University, Astana, Kazakhstan
- 6 March 2019 *The evolution of star clusters formed with a centrally peaked star formation efficiency*
Research Seminar at Physics Department, School of Science and Technology of Nazarbayev University, Astana, Kazakhstan
- 18 October 2018 *The life of star clusters, from birth to dissolution: a new approach*
Fesenkov Astrophysical Institute Seminar, Fesenkov Astrophysical Institute, Almaty, Kazakhstan
- 26 July 2018 *The life of star clusters, from birth to dissolution: a new approach*
ARI Institute Colloquium, Astronomisches Rechen-Institut, Heidelberg, Germany
- July 2018 *The life of open clusters: from formation until dissolution*
SFB-Cluster Meeting, Astronomisches Rechen-Institut, Heidelberg, Germany
- April 2018 *Bound fraction of clusters in different environment*
Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany
- February 2018 *Improving star cluster survival with centrally peaked star-formation efficiency profile*
ITA "blackboard" Colloquium, Institut für Theoretische Astrophysik, Heidelberg, Germany
- September 2017 *Progress discussion on open clusters after Violent Relaxation*
Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany
- July 2017 *Open Cluster Structure after Violent Relaxation*
Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany
- June 2017 *The impact of a star-formation efficiency profile on the evolution of open clusters*
SFB Seminar – The Milky Way System, Astronomisches Rechen-Institut, Heidelberg, Germany
- July 2016 *Improving The Survival Likelihood Of Star Clusters After Instantaneous Gas Expulsion*
Institute Seminar, Fesenkov Astrophysical Institute, Almaty, Kazakhstan
- February 2016 *STARDISK Project: The effect of the accretion disk on the central star cluster dynamics*

in AGNs

ARI Institute Kolloquium, Astronomisches Rechen-Institut, Heidelberg, Germany

January 2016

STARDISK Project: The effect of the accretion disk on the central star cluster dynamics in AGNs

Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany

April 2015

STARDISK: Impact of the accretion disk on the stellar phase space distribution function

Teeminar Heidelberg, Astronomisches Rechen-Institut, Heidelberg, Germany

PUBLICATIONS

First and/or corresponding author publications:

1. Assilhan, A., Mac Low, M-M., Polak, B., Abdikamalov, E., Cournoyer-Cloutier, C., Lewis, S. C., Kalambay, M., Otebay, A., **Shukirgaliyev, B. (2025)**. *Centrally concentrated star formation in young clusters*. *Astronomy & Astrophysics*, 705, A79. doi: [10.1051/0004-6361/202556268](https://doi.org/10.1051/0004-6361/202556268) [Q1]
2. Bissekenov, A., Kalambay, M., Abdikamalov, E., Pang, X., Berczik, P., **Shukirgaliyev, B. (2024)**. *Cluster membership analysis with supervised learning and N-body simulations*. *Astronomy & Astrophysics*, 689, A282. doi: [10.1051/0004-6361/202449791](https://doi.org/10.1051/0004-6361/202449791) [Q1]
3. **Shukirgaliyev, B.**, Otebay, A., Sobolenko, M. et al. (2021). *The bound mass of Dehnen models with centrally peaked star formation efficiency*. *Astronomy & Astrophysics*, 654, A53. doi: [10.1051/0004-6361/202141299](https://doi.org/10.1051/0004-6361/202141299) [Q1]
4. **Shukirgaliyev, B.**, Parmentier, G., Berczik, P., Just, A. (2020). *Star Clusters in the Galactic tidal field, from birth to dissolution*. *Star Clusters: From the Milky Way to the Early Universe*. *Proceedings of the International Astronomical Union*, Volume 351, pp. 507-511. doi: [10.1017/S1743921319006781](https://doi.org/10.1017/S1743921319006781) [Q4]
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7. **Shukirgaliyev, B.**, Parmentier, G., Just, A., Berczik, P. (2018). *The long-term evolution of star clusters formed with a centrally-peaked star-formation-efficiency profile*. *The Astrophysical Journal*, 863(2), 171. doi: [10.3847/1538-4357/aad3bf](https://doi.org/10.3847/1538-4357/aad3bf) [Q1]
8. Panamarev, T., **Shukirgaliyev, B.**, Meiron, Y. et al. (2018). *Star-disc interaction in galactic nuclei: formation of a central stellar disc*. *Monthly Notices of the Royal Astronomical Society*, 476, 4224. doi: [10.1093/mnras/sty459](https://doi.org/10.1093/mnras/sty459) [Q1]
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11. **Shukirgaliyev, B.**, Panamarev, T., Naurzbaeva, A. et al. (2016). *Effect of Gas Accretion Disc Profile on Orbital Parameters of the Accreted Stars*. *Reports of the National Academy of Sciences of the Republic of Kazakhstan*, Vol. 5, No. 309, p. 5-13.
12. **Shukirgaliyev, B.** (2016). *The effect of gaseous accretion disk on dynamics of the stellar cluster in AGN*. *Star Clusters and Black Holes in Galaxies across Cosmic Time*, *Proceedings of the International Astronomical Union, IAU Symposium*, Volume 312, pp. 113-117. doi: [10.1017/S1743921315007644](https://doi.org/10.1017/S1743921315007644) [Q4]
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14. Minglibayev, M. and **Shukirgaliyev, B.** (2011). *Equations of secular perturbations in the generalized two-body problem with variable mass*. *Bulletin of the Karaganda University*, Volume 1(61), pp. 73-77. <https://mathematics-vestnik.ksu.kz/apart/srch/2011.mathematics.1.61.2011.pdf>

Co-authored publications:

13. Bissekenov, A., Pang, X., Spurzem, R., **Shukirgaliyev, B.**, Kalambay, M., & Berczik, P. (2026) *Formation and Disruption of Wide Binaries in Star Clusters Revealed by N-body Simulations*. *The Astrophysical Journal*, 1001, L33. doi: [10.3847/2041-8213/ae5819](https://doi.org/10.3847/2041-8213/ae5819) [Q1, top decile]
14. Boumali, A., Jafari, N., **Shukirgaliyev, B.**, and Serdouk, F. (2025) *Thermal properties of Klein-Gordon oscillator in the context of Amelino-Camelia and Magueijo-Smolín doubly special relativity (DSR) frameworks* *European Physical Journal C*. 85(10), 1147. doi: [10.1140/epjc/s10052-025-14892-6](https://doi.org/10.1140/epjc/s10052-025-14892-6) [Q1, top decile]
15. Weis L., West C., Just A., Berczik P., Ishchenko M., Röser S., Schilbach E., **Shukirgaliyev B.** (2025) *Dynamical model of Praesepe and its tidal tails*. *Astronomy & Astrophysics*, 703, A47. doi: [10.1051/0004-6361/202451650](https://doi.org/10.1051/0004-6361/202451650) [Q1]
16. Bissekenov, A., Pang, X., Kamlah, A., Kouwenhoven, M. B. N., Spurzem, R., **Shukirgaliyev, B.**, Giersz, M., Askar, A., & Berczik, P. (2025) *Evolution of star clusters with initial bulk rotation via N-body simulations*. *Astronomy and Astrophysics*, 699, A196. doi: [10.1051/0004-6361/202554093](https://doi.org/10.1051/0004-6361/202554093) [Q1]
17. Ishchenko, M., Masliukh, V., Hradov, M., Berczik, P., **Shukirgaliyev, B.**, and Omarov, C. (2025) *Dynamical evolution of the open clusters with different star formation efficiencies and orbital parameters*. *Astronomy and Astrophysics*, 694, A33. doi: [10.1051/0004-6361/202452336](https://doi.org/10.1051/0004-6361/202452336) [Q1]
18. Askar, T., Lukac, M., **Shukirgaliyev, B.** and Abdikamalov, E., (2025). *Evaluating multi-GPU computing capabilities of Numba and CuPy*. *Cluster Computing*, 28(11), p.693. [Q1]

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21. Komesch T., Garay G., Henkel C., Omar A., Estalella R., Assembay Z., Li D., Guzmán A., Esimbek J., Huang J., He Y., Alimgazinova N., Kyzgarina M., **Shukirgaliyev B.**, Zhumabay N., Manapbayeva A. (2024) *Infall Motions in the Hot Core Associated with the Hypercompact H ii Region G345.0061+01.794 B*. *Astrophysical Journal*, 967 (1), 15. doi: [10.3847/1538-4357/ad3e7b](https://doi.org/10.3847/1538-4357/ad3e7b) [Q1]
22. Jafari, N. and **Shukirgaliyev, B.** (2024) *Nonrelativistic limits of the Klein-Gordon and Dirac equations in the Amelino-Camelia DSR*, *Physics Letters B*, 853, 138693. doi: [10.1016/j.physletb.2024.138693](https://doi.org/10.1016/j.physletb.2024.138693) [Q1]
23. Ussipov N., Akhmetali A., Zaidyn M., Akniyazova A., Sakan A., Kalambay M. and **Shukirgaliyev, B.** (2024) *Fractal dimension of star clusters*, *Eurasian Physical Technical Journal*, 21(3(49)), 108–116. doi: [10.31489/2024No3/108-116](https://doi.org/10.31489/2024No3/108-116) [Q3]
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29. Kalambay, M. T., Naurzbayeva, A. Z., Otebay, A. B., Abdinassilim, A. T., Kuvatova, D., Assilkhan, A. D., Panamarev, T., **Shukirgaliyev, B.**, and Berczik, P. P. (2022) *Mock observations of simulated star cluster on solar orbit* *Recent Contributions to Physics*, 83, 4-12. doi: [10.26577/RCPh.2022.v83.i4.01](https://doi.org/10.26577/RCPh.2022.v83.i4.01)

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31. Ishchenko, M., Sobolenko, M., Kalambay, M., **Shukirgaliyev, B.**, Berczik, P. (2021). *Milky Way Globular Clusters: close encounter rates with each other and with the Central Supermassive Black Hole* Reports of National academy of sciences of the Republic of Kazakhstan, 6(340), 94. doi: [10.32014/2021.2518-1483.116](https://doi.org/10.32014/2021.2518-1483.116)
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35. Kennedy, G., Meiron, Y., **Shukirgaliyev, B.** et al. (2016). *Star-disc interaction in galactic nuclei: orbits and rates of accreted stars*. *Monthly Notices of the Royal Astronomical Society*, 460, 240. doi: [10.1093/mnras/stw908](https://doi.org/10.1093/mnras/stw908) [Q1]

INFORMATION ABOUT HPC FACILITIES

at Jülich Supercomputing Center (Jülich, Germany)

JUWELS

as used in 2019 - 2020
 56 accelerated compute nodes
 2x Intel Xeon Gold 6148 CPU, 2x 20 cores, 2.4 GHz
 4x NVIDIA V100 GPU, 5120 cores, 16 GB HBM

JURECA

as used in 2016-2020
 75 accelerated compute nodes
 2x Intel Xeon E5-2680 v3 Haswell CPU, 2x 12 cores, 2.5 GHz
 2x NVIDIA K80 GPUs, 4992 cores, 24 GB GDDR5

*MilkyWay
de-installed*

as used in 2011 - 2015
 206 Compute nodes IBM System x iDataPlex dx360 M3
 2x Intel Xeon X5650(Westmere) 6-core CPU 2.66 GHz
 2x NVIDIA Tesla M2050 (Fermi) 1.15 GHz (448 cores), 3 GB (on 54 Nodes)
 2x NVIDIA Tesla M2070 (Fermi) 1.15 GHz (448 cores), 6 GB (on 152 Nodes)

at Heidelberg University (Heidelberg, Germany)

*bwForCluster
MLS&WISO*

as used in 2018
 48 nodes with Nvidia Tesla K80 (Kepler) 4992 cores, 24 GB

Kepler

as used in 2013 - 2020
 12 nodes with Nvidia Tesla K20 (Kepler) 2496 cores, 5 GB

at National Observatories of China (Beijing, China)

Laohu

as used in 2013 - 2018
 85 nodes with Nvidia Tesla K20 (Kepler) 2496 cores, 5 GB

at Nazarbayev University (Astana, Kazakhstan)

Shabyt

as used in 2020 - 2023
 4 accelerated nodes 2x AMD EPYC 7452 CPUs (32 cores / 64 threads, 2.3 MHz Base),
 2x NVidia Tesla V100 GPU, 5120 cores, 32GB HBM2 RAM

DGX2

as used in 2020 - 2023
 1 accelerated node with 2x Intel Xeon Platinum 8168, 2.7 GHz, 24-cores
 16x NVidia Tesla V100 GPU, 5120 cores, 32GB HBM2

*ECL computing
facilities*

as used in 2020 - 2025
 1 accelerated node with
 2x AMD EPYC 9754, 256-cores,
 2x Nvidia RTX A4000 6144 cores, 16GB

1 accelerated node with
 2x AMD EPYC 7763, 128-cores,
 2x Nvidia RTX A4000 6144 cores, 16GB

4 accelerated nodes with
 1x AMD Ryzen Threadripper 3990X, 2.9GHz, 64-cores
 2x NVidia GeForce RTX 3090 GPU, 10496 cores, 24GB (3 nodes)
 2x NVidia GeForce RTX 3080 GPU, 8704 cores, 10GB (1 nodes)

*FAI computing
facilities*

at Fesenkov Astrophysical Institute (Almaty, Kazakhstan)

as used in 2020 - 2022
 4 accelerated nodes with 2x NVidia GeForce RTX 3090 GPU, 10496 cores, 24GB

September 8, 2026